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Project – Low Level Design
on

Cloud Cost Intelligence Platform for Smart City

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1. Introduction

1.1 Purpose

This Low Level Design (LLD) document provides a detailed, implementation-level description of the CloudLens application. It expands upon the High Level Design (HLD) by specifying exact function signatures, class structures, variable names, algorithm logic, data validation rules, and inter-module interactions as reflected directly in the source code (app.py).

1.2 Scope

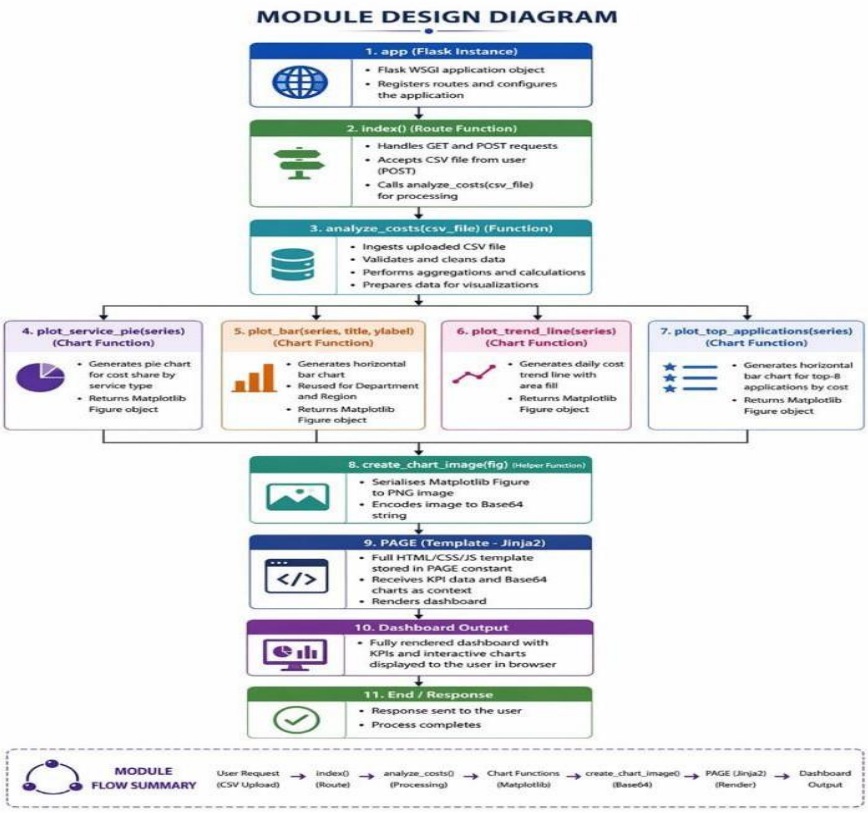
The LLD covers the following modules present in the single-file Flask application:

- Flask route handler (index)
- Data ingestion and analysis module (analyze_costs)
- Chart generation module — five dedicated plot functions
- Chart serialisation helper (create_chart_image)
- Jinja2 HTML/CSS template (PAGE constant)

1.3 Definitions and Abbreviations

Term	Definition
CSV	Comma-Separated Values — the input file format accepted by the application
DataFrame (df)	In-memory tabular data structure provided by the pandas library
Base64	Binary-to-text encoding used to embed PNG chart images inline in HTML
BytesIO	Python in-memory byte stream; used to avoid writing chart files to disk
DXA	Unit used in DOCX formatting (1 inch = 1440 DXA)
KPI	Key Performance Indicator — summary metric shown in the dashboard cards
INR	Indian Rupee — the cost unit used throughout the data model
MVC	Model-View-Controller — the architectural pattern used by Flask
Jinja2	Python templating engine integrated with Flask

2. Module Design



Module / Symbol	Type	Responsibility
app	Flask instance	WSGI application object; registers routes
PAGE	str constant	Full HTML/CSS/JS template rendered by Jinja2
index()	Route function	Handles GET and POST requests; orchestrates analysis and rendering
analyze_costs(csv_file)	Function	Ingests, validates, and aggregates the uploaded CSV
create_chart_image(fig)	Helper function	Serialises a Matplotlib Figure to a base64 PNG string
plot_service_pie(series)	Chart function	Generates a pie chart of cost share by service type
plot_bar(series, title, ylabel)	Chart function	Generic horizontal bar chart (reused for Department and Region)
plot_trend_line(series)	Chart function	Generates the daily cost trend line with area fill
plot_top_applications(se ries)	Chart function	Horizontal bar chart for top-8 applications by cost
_fig(w, h)	Helper function	Factory for styled Matplotlib (fig, ax) tuples with dark theme

3. Detailed Function Specifications

3.1 index()

3.1.1 Signature

```
def index() -> flask.Response
```

3.1.2 Route Binding

```
@app.route("/", methods=["GET", "POST"])
```

3.1.3 Local Variables

Variable	Type	Initial Value	Purpose
filename	str | None	None	Stores the original filename of the uploaded CSV for display
error	str | None	None	Holds a user-facing error message if processing fails
kpis	dict | None	None	Formatted KPI strings passed to the Jinja2 template
charts	dict	{}	Maps chart names to base64 PNG strings for template embedding

3.1.4 Logic Flow

- GET request: returns render_template_string(PAGE) with all optional variables set to None / empty dict — renders blank dashboard.
- POST request:
  - Retrieves file from request.files.get('myfile').
  - If no file or filename is empty: sets error = 'No file chosen.' and falls through to render.
  - Otherwise: calls analyze_costs(file), populates kpis dict with formatted strings (f'{value:,}'), generates five charts by calling each plot function and passing the result to create_chart_image().
  - Any Exception raised inside the try block is caught; error is set to f'Could not analyse file — {exc}'.
- Returns render_template_string(PAGE, filename=filename, error=error, kpis=kpis, charts=charts).

3.1.5 KPI Dictionary Schema

Key	Source Expression	Format
total_cost	analysis['total_cost']	f'{value:,}' — comma-separated integer string
avg_daily_cost	analysis['avg_daily_cost']	f'{value:,}'
top_department_name	analysis['top_department_name']	Plain string
top_department_cost	analysis['top_department_cost']	f'{value:,}'
top_application_name	analysis['top_application_name']	Plain string
top_application_cost	analysis['top_application_cost']	f'{value:,}'

3.2 analyze_costs(csv_file)

3.2.1 Signature

```
def analyze_costs(csv_file: FileStorage) -> dict
```

3.2.2 Input

csv_file — a Werkzeug FileStorage object (file-like stream) received from request.files.

3.2.3 Validation Rules

Check	Method	Exception Raised
'Date' column present	Column membership check	ValueError: 'CSV must include a Date column.'
Date values parseable	pd.to_datetime(errors='coerce') + isna().all()	ValueError: 'Could not parse any Date values.'
'Cost_INR' column present	Column membership check	ValueError: 'CSV must include a Cost_INR column.'
Required categorical columns present	df.dropna(subset=[...])	Silently drops incomplete rows; no exception

3.2.4 Processing Steps (in order)

- Read CSV: df = pd.read_csv(csv_file)
- Parse dates: df['Date'] = pd.to_datetime(df['Date'], errors='coerce')
- Coerce costs: df['Cost_INR'] = pd.to_numeric(df['Cost_INR'], errors='coerce').fillna(0)

- Drop nulls: df = df.dropna(subset=['Department', 'Service_Type', 'Application', 'Region', 'Date'])
- Compute aggregations — see table below.

3.2.5 Aggregations

Output Key	Pandas Expression	Type
total_cost	int(df['Cost_INR'].sum())	int
avg_daily_cost	int(df.groupby('Date')['Cost_INR'].sum().mean())	int
top_department_name	department_cost.index[0]	str
top_department_cost	int(department_cost.iloc[0])	int
top_application_name	top_application_costs.index[0]	str
top_application_cost	int(top_application_costs.iloc[0])	int
department_cost	df.groupby('Department')['Cost_INR'].sum().sort_values(ascending=False)	pd.Series
region_cost	df.groupby('Region')['Cost_INR'].sum().sort_values(ascending=False)	pd.Series
service_costs	df.groupby('Service_Type')['Cost_INR'].sum().sort_values(ascending=False)	pd.Series
cost_trend	df.groupby('Date')['Cost_INR'].sum().sort_index()	pd.Series (DatetimeIndex)
top_application_costs	df.groupby('Application')['Cost_INR'].sum().sort_values(ascending=False).head(8)	pd.Series

3.3 create_chart_image(fig)

3.3.1 Signature

```
def create_chart_image(fig: matplotlib.figure.Figure) -> str
```

3.3.2 Algorithm

- Create an in-memory buffer: buf = BytesIO()
- Save the figure: fig.savefig(buf, format='png', bbox_inches='tight', facecolor=BG)
- Seek to start: buf.seek(0)
- Encode: data = base64.b64encode(buf.read()).decode('utf-8')
- Close figure to free memory: plt.close(fig)
- Return data — the caller embeds it as src="{data}" in the HTML <img> tag.

3.4 _fig(w=6, h=4.2)

3.4.1 Signature

```
def _fig(w: float = 6, h: float = 4.2) -> tuple[Figure, Axes]
```

3.4.2 Behaviour

Factory function that creates a consistently styled (fig, ax) pair. Applies the dark theme by setting figure and axes background colours, spine colours, tick label colours, and grid style using the module-level colour constants.

Constant	Hex Value	Applied To
BG	#08090b	Figure and axes facecolor; chart image background
SURFACE	#0e1117	Axes facecolor (plot area)
TEXT	#e8eaf0	Tick labels, axis titles, chart titles
MUTED	#6b7280	Axis label text, tick mark colours
GRID	#1a1e28	Grid lines and spine border colour

3.5 Chart Generation Functions

3.5.1 plot_service_pie(series)

Attribute	Value
Input	pd.Series — index: service type labels; values: cost totals
Figure size	5 × 4.2 inches, 110 DPI (created independently, not via _fig())
Plot type	Pie chart via ax.pie()
Palette	['#5b8af7','#34d399','#f59e0b','#a78bfa','#f87171','#22d3ee','#fb923c','#e879f9']
wedgeprops	linewidth=1.2, edgecolor=BG
pctdistance	0.78 (percentage labels placed at 78% of radius)
startangle	140 degrees
autopct	'%1.1f%%'
Title	'Service share'

3.5.2 plot_bar(series, title, ylabel)

Attribute	Value
Input	pd.Series (sorted descending); title: str; ylabel: str
Figure size	6 × 4.2 inches via _fig()
Plot type	Horizontal bar chart via ax.barh() — series reversed for top-to-bottom display
Colour scheme	Top bar: '#5b8af7' (accent blue); remaining bars: '#2d4a8a' (dark blue)
Bar height	0.62
Grid	x-axis only (color=GRID, linewidth=0.6); y-axis grid hidden
Used for	Department breakdown and Regional distribution (both reuse this function)

3.5.3 plot_trend_line(series)

Attribute	Value
Input	pd.Series with DatetimeIndex and daily cost values
Figure size	7 × 4 inches via _fig()
Plot type	Line plot + area fill + scatter markers
Line	color='#5b8af7', linewidth=1.8, solid_capstyle='round'
Fill	ax.fill_between() with alpha=0.12, color='#5b8af7'
Markers	ax.scatter() with s=28, edgecolors=BG, linewidths=1.2, zorder=5
Date formatting	fig.autofmt_xdate(rotation=30)

3.5.4 plot_top_applications(series)

Attribute	Value
Input	pd.Series — top 8 applications by cost (pre-sliced in analyze_costs)
Figure size	6 × 4.2 inches via _fig()
Plot type	Horizontal bar chart via ax.barh()
Colour scheme	Top bar: '#34d399' (green); remaining: '#1a6649' (dark green)
Bar height	0.62

4. Data Flow — Request Lifecycle

The following table traces every transformation a data point undergoes from the moment the CSV is submitted to the rendered HTML response.

Steps	Operation	Input Type	Output Type	Key Function / Library
1	Browser submits form	HTML multipart/form-data	HTTP POST body	Browser
2	Flask receives file stream	HTTP POST body	FileStorage object	flask.request.files
3	CSV parsed into DataFrame	FileStorage stream	pd.DataFrame	pd.read_csv()
4	Date column parsed	String column	datetime64 column	pd.to_datetime()
5	Cost column coerced	Mixed string/numeric	float64 column	pd.to_numeric()
6	Null rows dropped	DataFrame with NaNs	Clean DataFrame	df.dropna()
7	Aggregation computed	Clean DataFrame	pd.Series objects	df.groupby().sum()
8	KPI values extracted	pd.Series	int / str scalars	series.index[0], series.iloc[0]
9	Matplotlib figure created	pd.Series	Figure object	plot_* functions + _fig()
10	Figure saved to buffer	Figure object	PNG bytes (BytesIO)	fig.savefig()
11	Buffer encoded	PNG bytes	base64 str	base64.b64encode()
12	Template rendered	base64 str + KPI dict	HTML string	render_template_string()
13	HTML returned to browser	HTML string	HTTP 200 Response	Flask

5. Template Design (PAGE Constant)

5.1 Structure

The PAGE constant is a single multi-line Python string containing a complete HTML5 document. It is rendered by Jinja2 through render_template_string(). The template is divided into the following sections:

Template Section	Jinja2 Variables Used	Purpose
Top bar	None	Brand identity and application title
Page heading	None	Static title and description
Upload card	filename	File input form; displays last uploaded filename
Error block	error	Conditionally rendered; shown only when {% if error %}
KPI row (4 cards)	kpis.total_cost, kpis.avg_daily_cost, kpis.top_department_name, kpis.top_department_cost, kpis.top_application_name, kpis.top_application_cost	Displays key metric cards; shows '—' placeholder when kpis is None
Chart: Daily Trend	charts.trend_line	Wide (span-8) line chart; renders <img> with base64 src or empty-state div
Chart: Service Pie	charts.service_pie	Narrow (span-4) pie chart
Chart: Department Bar	charts.department_bar	Narrow (span-4) horizontal bar
Chart: Region Bar	charts.region_bar	Narrow (span-4) horizontal bar
Chart: Top Apps	charts.top_app_bar	Narrow (span-4) horizontal bar
Inline JavaScript	None	updateName() — updates filename label on file selection

5.2 CSS Variable Reference

CSS Variable	Value	Usage
--bg	#08090b	Page background; chart image background
--surface	#0e1117	Card and chart panel background
--surface2	#141821	Secondary surface (upload icon background)
--border	rgba(255,255,255,0.07)	Default card/panel border
--border-hi	rgba(255,255,255,0.13)	Upload zone and icon borders
--accent	#5b8af7	Primary interactive colour (button, file label, charts)
--accent-dim	rgba(91,138,247,0.12)	Upload zone hover background
--green	#34d399	KPI trend label colour; top-apps chart
--amber	#f59e0b	Top department card accent
--red	#f87171	Error message colour
--muted	#6b7280	Secondary text, axis labels
--mono	'DM Mono', monospace	Monospace font for KPI labels, file names, sub-labels
--sans	'Sora', sans-serif	Primary UI font
--radius	14px	Default card border radius
--radius-lg	22px	Large card border radius (upload card, KPI cards, chart cards)

5.3 Responsive Breakpoints

Breakpoint	CSS Change
max-width: 1024px	KPI row switches to 2-column grid; all chart cards expand to full-width (span 12)
max-width: 640px	Shell padding reduced; KPI row becomes 2×2 grid; upload card stacks vertically; top bar stacks vertically

6. Error Handling

Scenario	Trigger	Handling Mechanism	User-Facing Output
No file submitted	file is None or filename == ''	Explicit check in index()	error = 'No file chosen. Please pick a CSV.'
Missing Date column	'Date' not in df.columns	ValueError raised in analyze_costs()	error = 'Could not analyse file — CSV must include a Date column.'
Unparseable dates	df['Date'].isna().all()	ValueError raised in analyze_costs()	error = 'Could not analyse file — Could not parse any Date values.'
Missing Cost_INR column	'Cost_INR' not in df.columns	ValueError raised in analyze_costs()	error = 'Could not analyse file — CSV must include a Cost_INR column.'
Malformed cost values	Non-numeric Cost_INR entries	pd.to_numeric(errors='coerce').fillna(0)	Silently set to 0; no error shown
Missing categorical columns	NaN in Department, Service_Type, etc.	df.dropna() removes affected rows	Silently drops rows; analysis continues
Any other unexpected exception	Unhandled exception inside try block	except Exception as exc: error = f'Could not analyse file — {exc}'	Generic error message surfaced in UI

7. Module-Level Constants

Constant	Value	Type	Purpose
BG	'#08090b'	str	Darkest background colour for charts
SURFACE	'#0e1117'	str	Axes area colour for charts
TEXT	'#e8eaf0'	str	Light text colour for chart labels
MUTED	'#6b7280'	str	Muted grey for axis labels and secondary text
GRID	'#1a1e28'	str	Grid line and spine colour
PAGE	Multi-line HTML string (~450 lines)	str	Full Jinja2 template for the dashboard

8. Dependency Catalogue

Library	Import Alias	Version Constraint	Usage in Code
Flask	flask	>=2.0	Flask(), app.route(), request, render_template_string
pandas	pd	>=1.3	read_csv, to_datetime, to_numeric, groupby, dropna, Series
matplotlib	matplotlib / plt	>=3.4	Figure, Axes, pyplot; all chart generation and serialisation
numpy	np	>=1.21	Imported but not directly called in current code (transitive dependency of pandas/matplotlib)
base64	(stdlib)	N/A	b64encode() — encodes PNG bytes to ASCII string
io.BytesIO	(stdlib)	N/A	In-memory binary buffer for PNG serialisation

9. Testing Considerations

9.1 Unit Test Cases

Test ID	Function Under Test	Input	Expected Output
UT-01	analyze_costs()	Valid CSV with all required columns	Dict with correct aggregated values
UT-02	analyze_costs()	CSV missing 'Date' column	ValueError with message 'CSV must include a Date column.'
UT-03	analyze_costs()	CSV with all invalid dates	ValueError with message 'Could not parse any Date values.'
UT-04	analyze_costs()	CSV missing 'Cost_INR' column	ValueError with message 'CSV must include a Cost_INR column.'
UT-05	analyze_costs()	CSV with non-numeric Cost_INR values	Rows with bad costs set to 0; no exception
UT-06	create_chart_image()	Valid Matplotlib Figure	Non-empty base64-encoded string
UT-07	index() — GET	HTTP GET /	200 response; charts and kpis absent from context

Test ID	Function Under Test	Input	Expected Output
UT-08	index() — POST (valid)	HTTP POST with valid CSV	200 response; charts dict has 5 keys; kpis dict has 6 keys
UT-09	index() — POST (no file)	HTTP POST with no file	200 response; error message set

9.2 Sample CSV Structure

A minimal valid test CSV must include the following headers and representative data:

```
Date,Cost_INR,Department,Service_Type,Application,Region
2024-01-01,12500,Engineering,Compute,AuthService,ap-south-1
2024-01-01,8300,Marketing,Storage,CampaignApp,us-eas
2024-01-02,15700,Engineering,Network,APIGateway,ap-south-1
```

10. References

#	Reference	URL
1	Flask Documentation	https://flask.palletsprojects.com/
2	Pandas User Guide	https://pandas.pydata.org/docs/user_guide/
3	Matplotlib Documentation	https://matplotlib.org/stable/contents.html
4	Jinja2 Template Designer Docs	https://jinja.palletsprojects.com/en/3.x/templates/
5	Python base64 module	https://docs.python.org/3/library/base64.html
6	Python io.BytesIO	https://docs.python.org/3/library/io.html